

Enterprise Campus Network Segmentation Lab

A Cisco Packet Tracer simulation of an enterprise campus network with VLAN segmentation, 802.1Q trunking, hierarchical switching, and Router-on-a-Stick inter-VLAN routing across 17 departments.

CISCO PACKET TRACER

VLAN SEGMENTATION

ROUTER-ON-A-STICK

802.1Q

INTER-VLAN ROUTING

✓ LAB COMPLETE

KALI LINUX 2025.2

CISCO IOS 15.1

LAB COMPLETE – 0% PACKET LOSS

9

VLANS

18

DEPARTMENTS

9

SWITCHES

1

ROUTER
(ROAS)

0%

PACKET
LOSS

OVERVIEW

This project simulates a small enterprise campus network using Cisco Packet Tracer. The environment demonstrates foundational network engineering and cybersecurity concepts including VLAN segmentation, 802.1Q trunking, centralized switching architecture, and inter-VLAN routing using Router-on-a-Stick.

The lab evolved from an initially unstable flat topology into a redesigned hierarchical enterprise architecture with a dedicated core/distribution switch and multiple departmental access switches – mirroring real-world enterprise infrastructure design patterns.



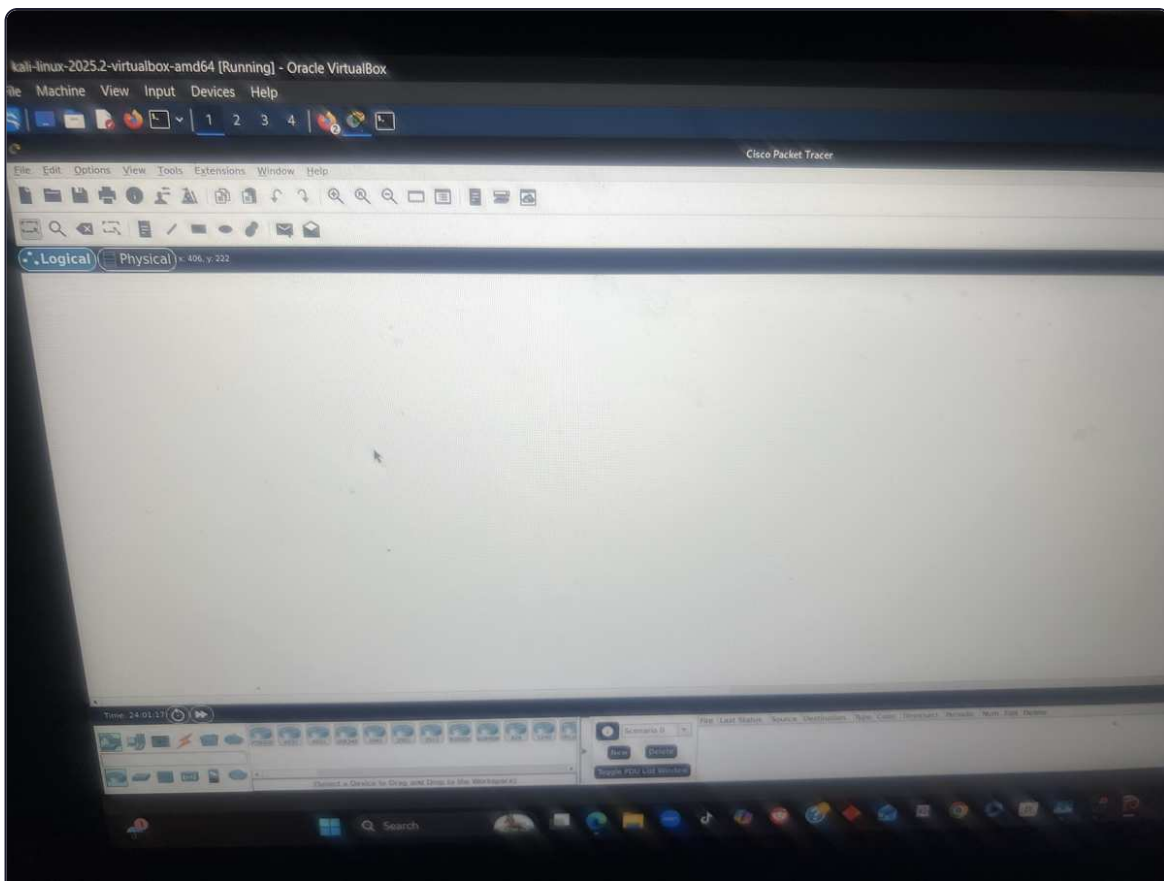
PRIMARY FILE

`master network.pkt` – Final completed topology • 18 departments • 9 switches • Router-on-a-Stick

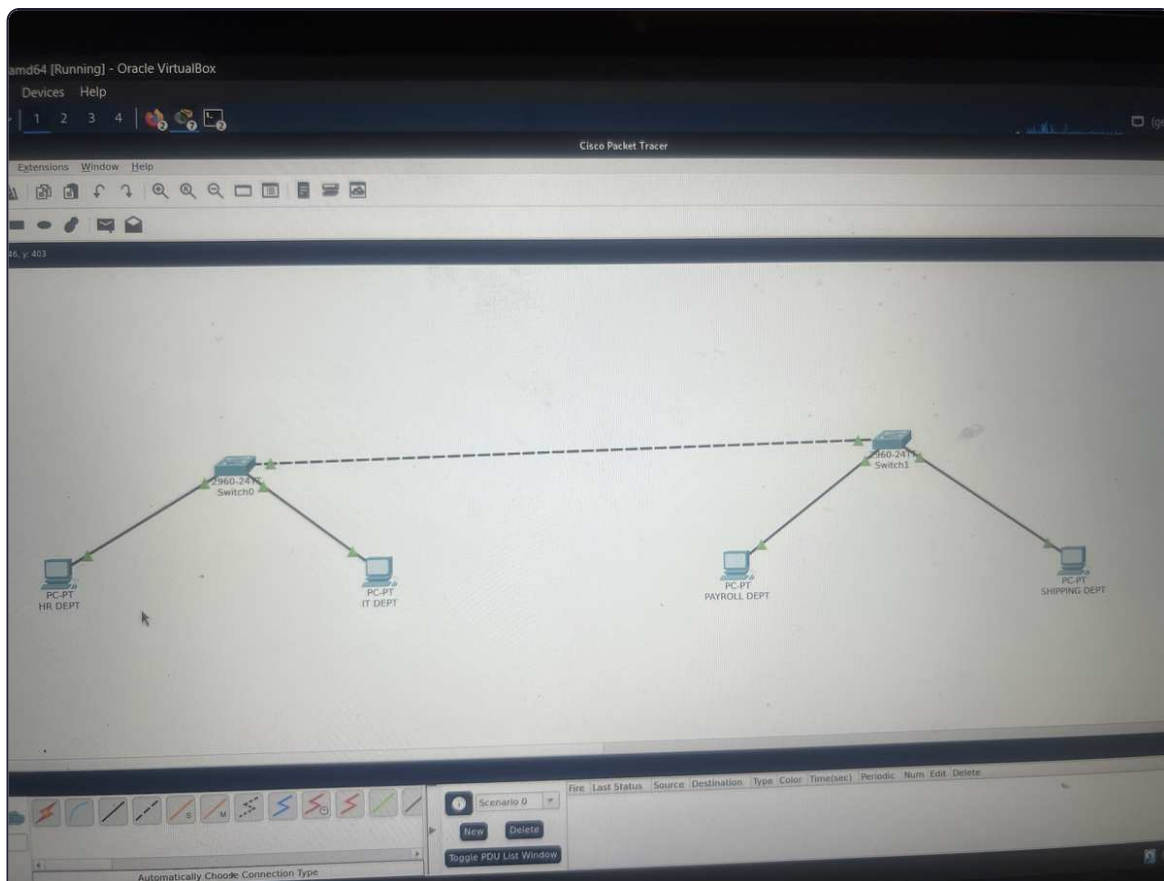
TOPOLOGY EVOLUTION

// Initial Topology – Flat Design

The lab started with a flat, unstructured topology that quickly proved unstable and unscalable as departments were added. Without a core switch, trunk and routing conflicts emerged at scale.



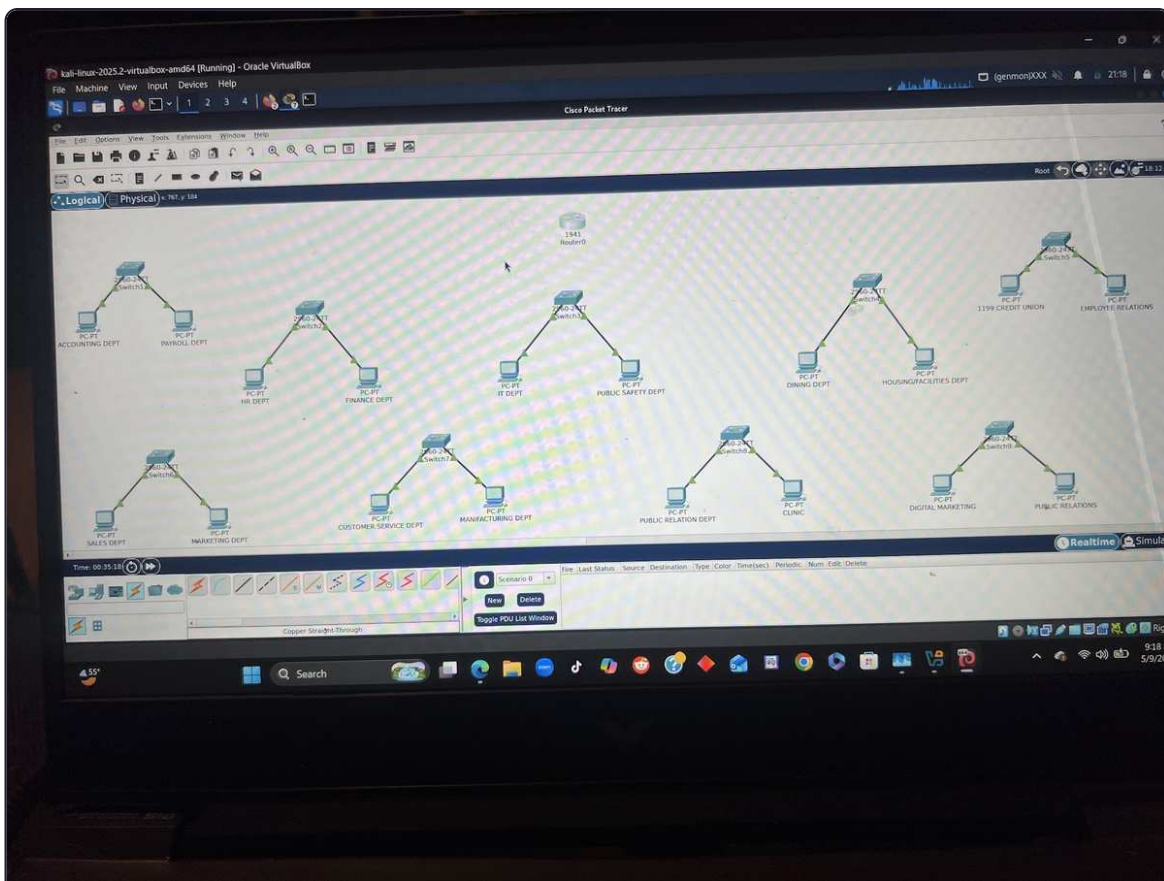
Initial topology – blank canvas before devices were added



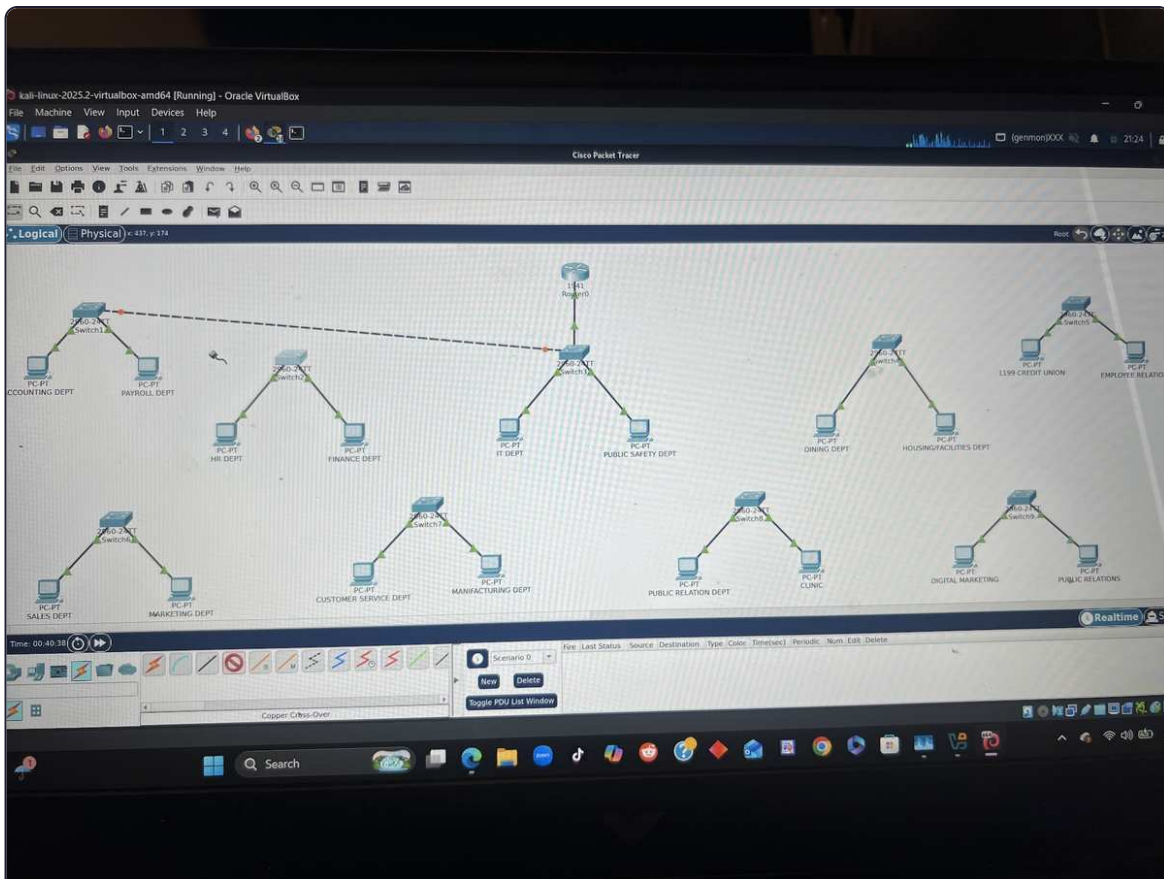
Initial topology – flat design before hierarchical redesign

// Redesigned Hierarchical Topology

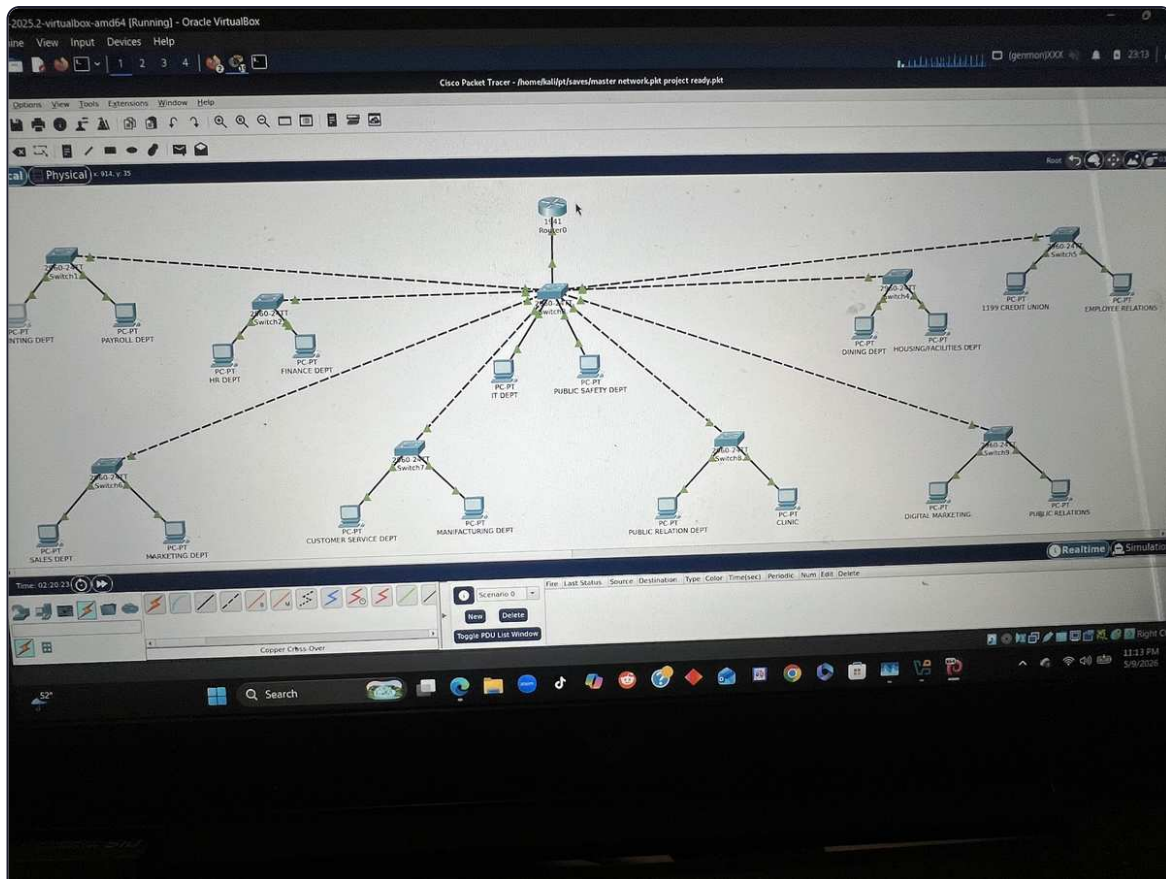
After identifying failures in the flat design, the topology was fully rebuilt using a hierarchical enterprise model with Switch3 as the centralized core/distribution layer.



Redesigned hierarchical topology with centralized core switch



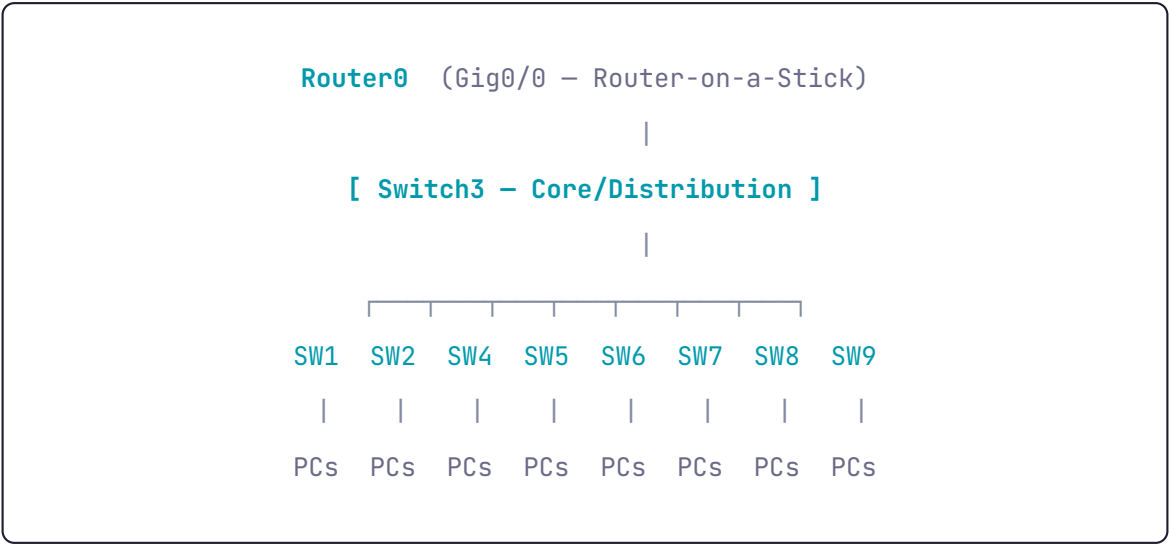
Final stable topology – all links green, all VLANs routed



master network.pkt project ready – final completed topology

NETWORK ARCHITECTURE

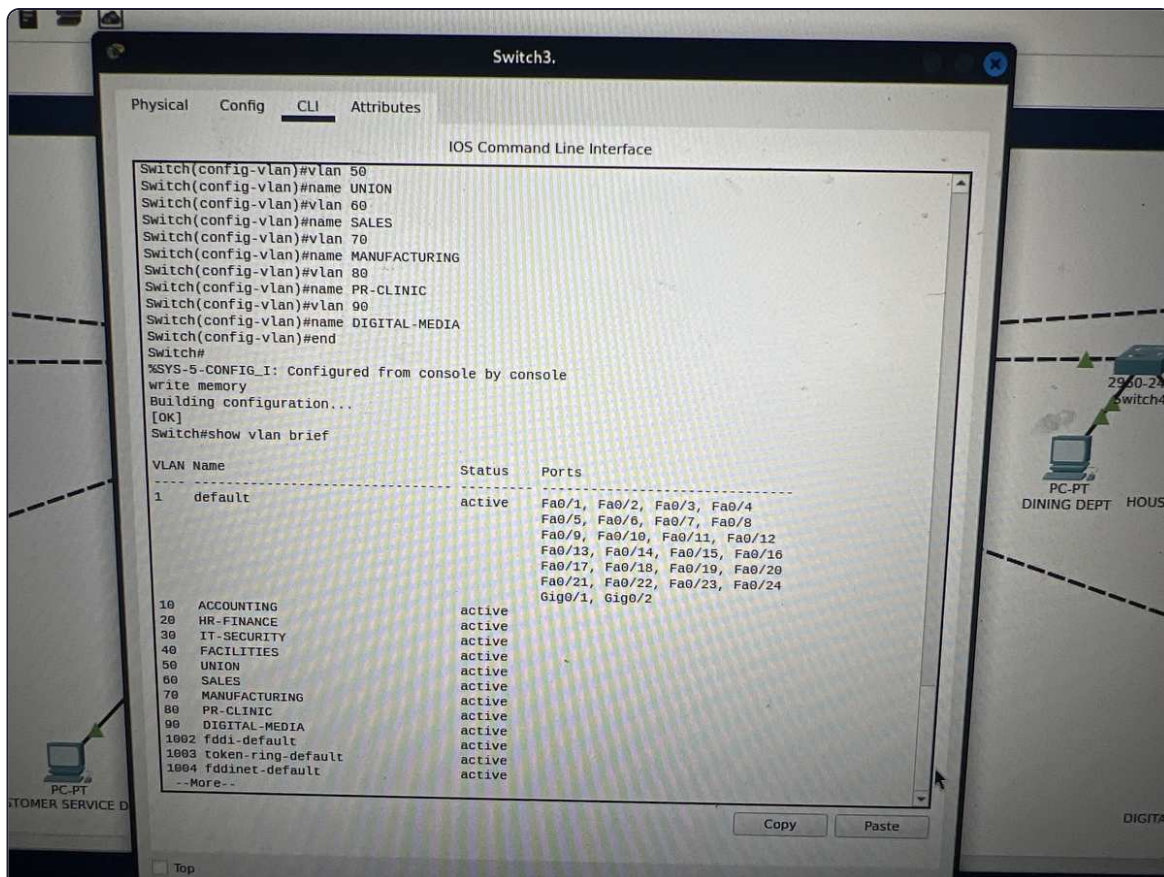
The topology follows a three-tier hierarchical design. Router0 sits at the top providing inter-VLAN routing. Switch3 acts as the core/distribution layer aggregating all trunk links. Nine access switches fan out to departmental PCs.



LAYER	ROLE	DEVICE
Routing	Inter-VLAN routing via subinterfaces	Router0
Core/Distribution	VLAN aggregation, trunk uplinks	Switch3
Access	Per-department ports, VLAN assignment	SW1, SW2, SW4-SW9

VLAN SEGMENTATION PLAN

Without VLANs, all departments share one broadcast domain. VLANs create logically separate networks on shared physical infrastructure – a compromised Accounting workstation should not have Layer 2 adjacency to HR or Finance.



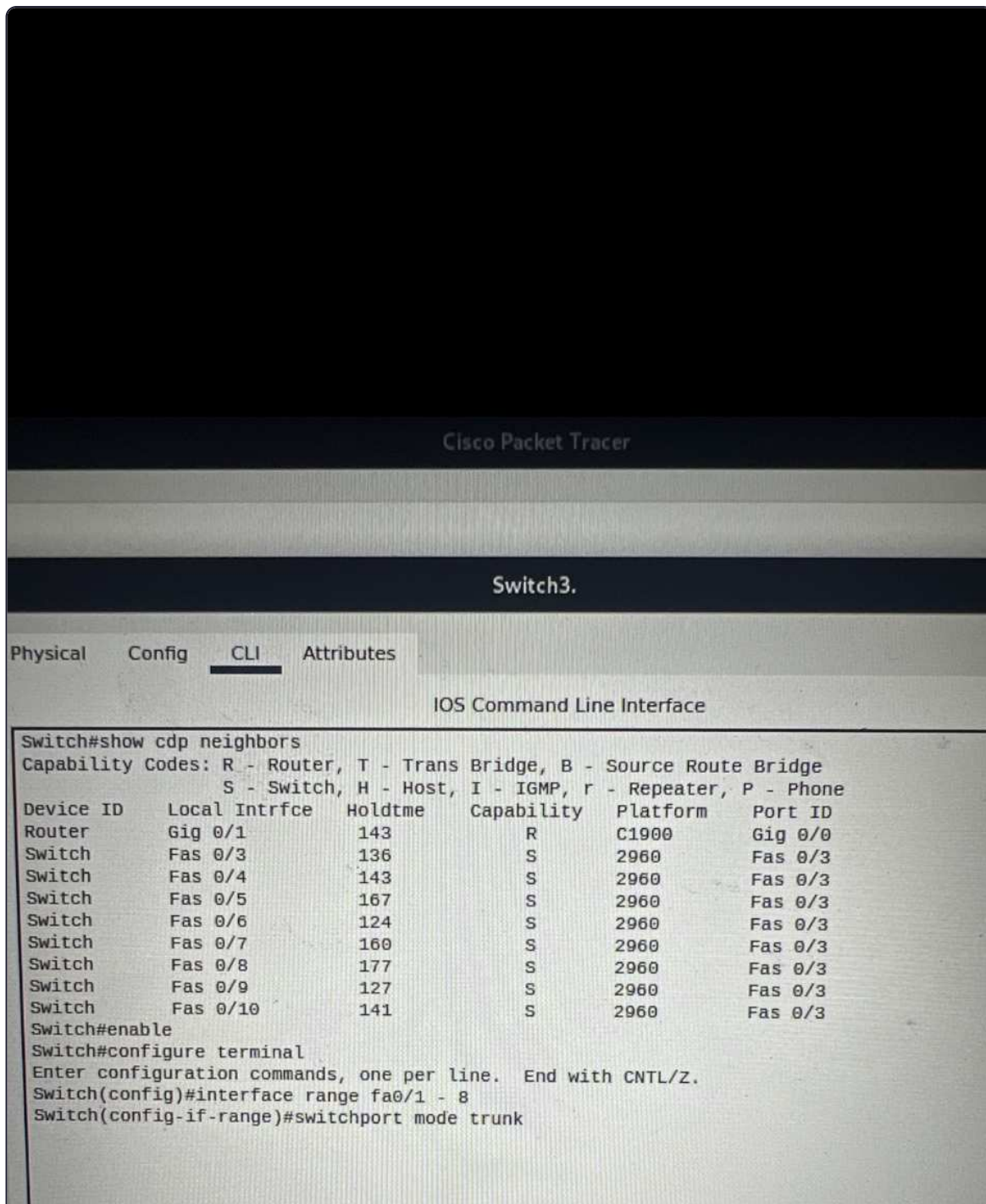
show vlan brief – all 9 VLANs created, named, and active on Switch3

VLAN	DEPARTMENT	SUBNET
10	Accounting / Payroll	192.168.10.0/24
20	HR / Finance	192.168.20.0/24
30	IT / Public Safety	192.168.30.0/24
40	Dining / Facilities	192.168.40.0/24
50	Union / Employee Relations	192.168.50.0/24
60	Sales / Marketing	192.168.60.0/24
70	Customer Service / Manufacturing	192.168.70.0/24
80	PR / Clinic	192.168.80.0/24
90	Digital Media / Public Relations	192.168.90.0/24

LAYER 2 – TRUNK CONFIGURATION

802.1Q tagging stamps each frame with its VLAN ID so the receiving device routes it correctly. Without trunks, you'd need a separate physical cable for every VLAN.

```
switchport mode trunk
```

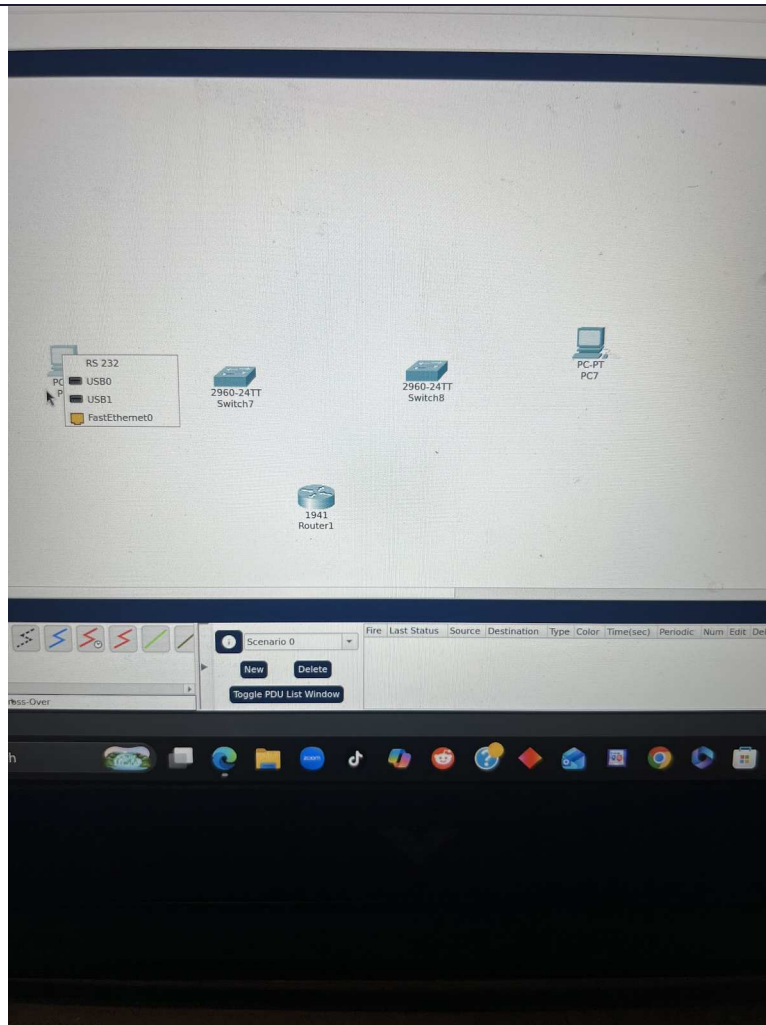


```
Switch(config-if-range)#  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
```

Copy

☐ Top

Trunk config applied – CDP neighbors confirmed on Switch3



All trunk interfaces stabilized to up/up after 802.1Q negotiation

LAYER 3 – ROUTER-ON-A-STICK

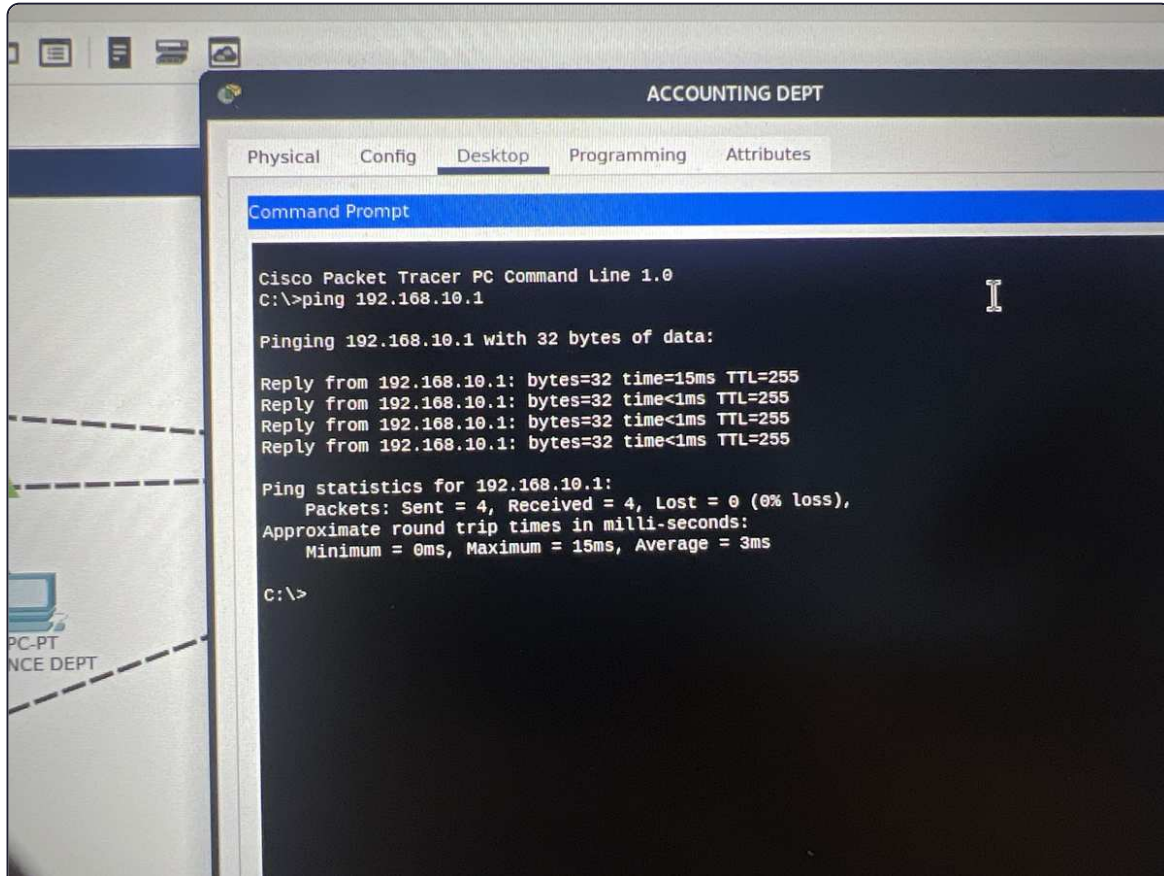
Router-on-a-Stick uses subinterfaces – logical divisions of one physical interface, each tagged with a different VLAN ID. Each subinterface acts as the default gateway for its VLAN.

```
interface gig0/0
  no shutdown

interface gig0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
```

```
interface gig0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
```

Proof of Functionality



4/4 replies - 0% packet loss - Accounting DEPT → 192.168.10.1 - inter-VLAN routing confirmed

TROUBLESHOOTING SCENARIOS

Real failures encountered during the build – not simulated errors.

// Subinterface VID Conflict

%Configuration of multiple subinterfaces of the same main interface with

VID CONFLICT

Root cause: VLAN 10 encapsulation assigned to wrong subinterface (gig0/0.20 instead of gig0/0.10).

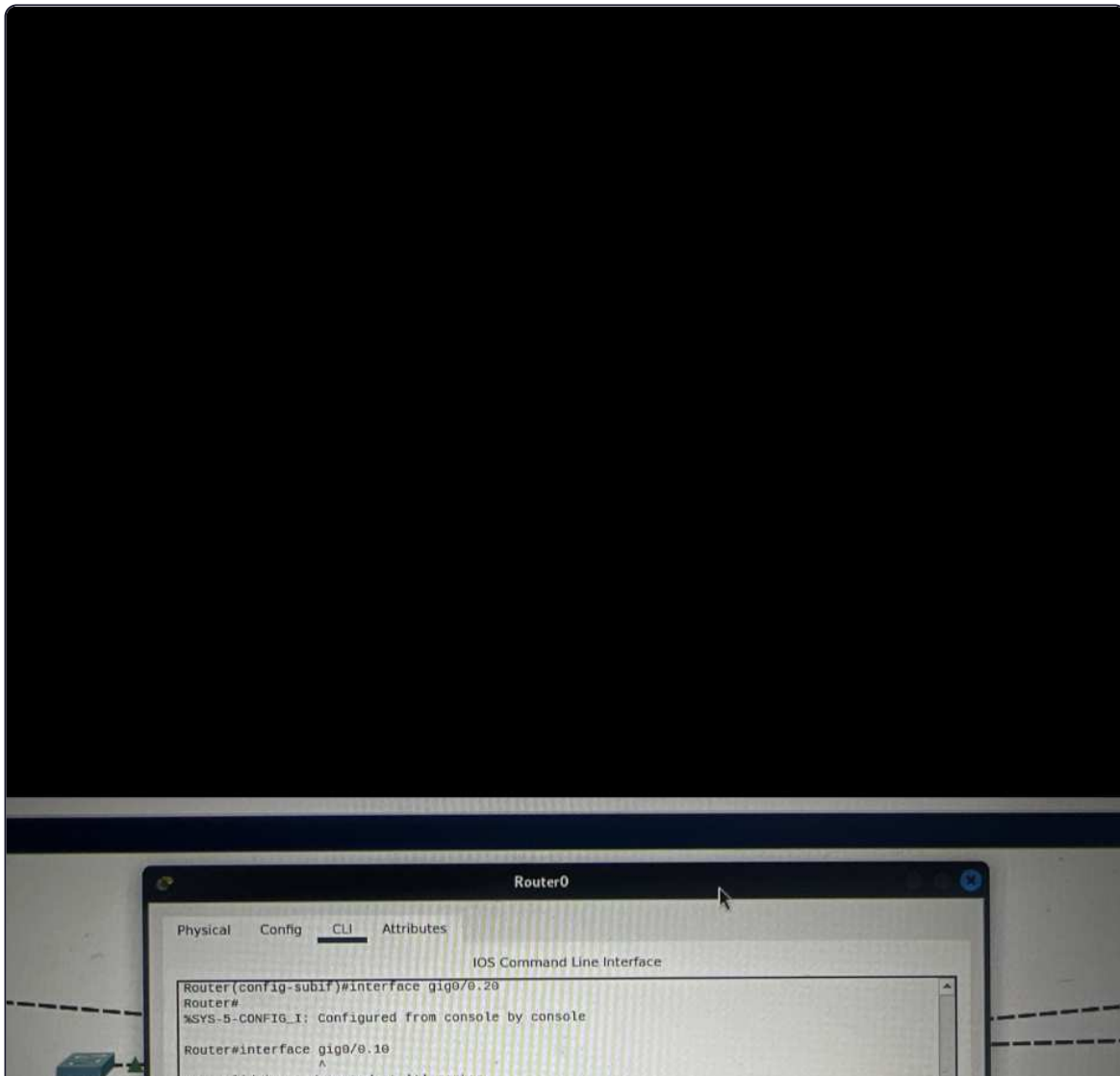
Fix: Re-entered `encapsulation dot1Q 10` on gig0/0.10 and `encapsulation dot1Q 20` on gig0/0.20.

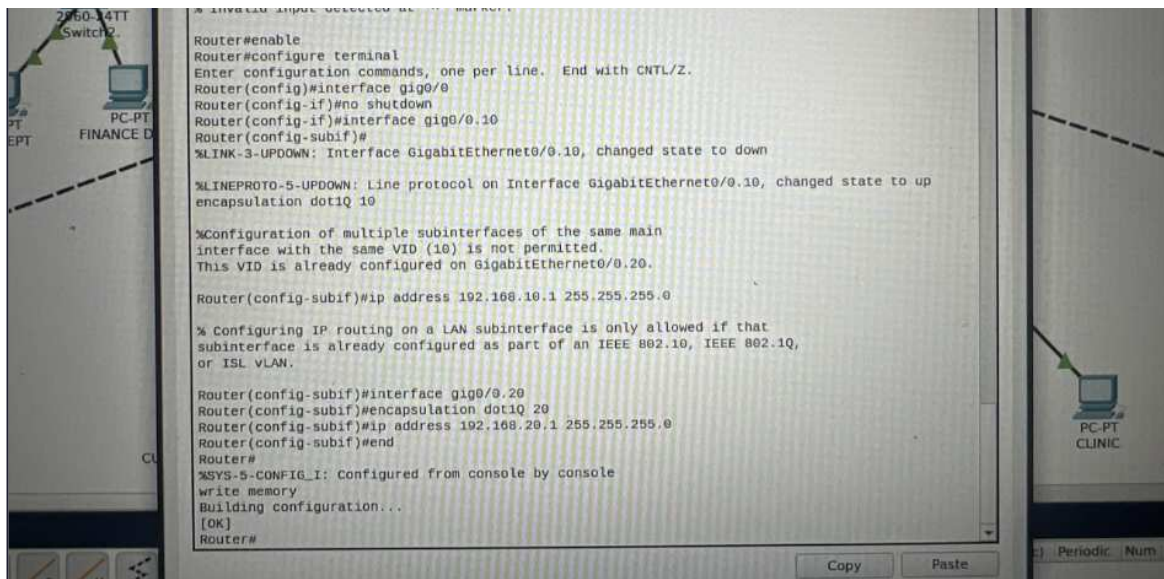
// IP Before Encapsulation

% Configuring IP routing on a LAN subinterface is only allowed if that su

ENCAPSULATION ORDER

`ip address` was entered before `encapsulation dot1Q`. Always configure encapsulation first.





Router0 CLI – VID conflict and encapsulation order errors captured live

// Trunk Negotiation / Interface Flapping

```
%LINEPROTO-5-UPDOWN: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Interface FastEthernet0/1, changed state to up
```

DTP FLAP

Expected DTP negotiation behavior. All interfaces stabilized to up/up.

// Native VLAN Mismatch

NATIVE VLAN

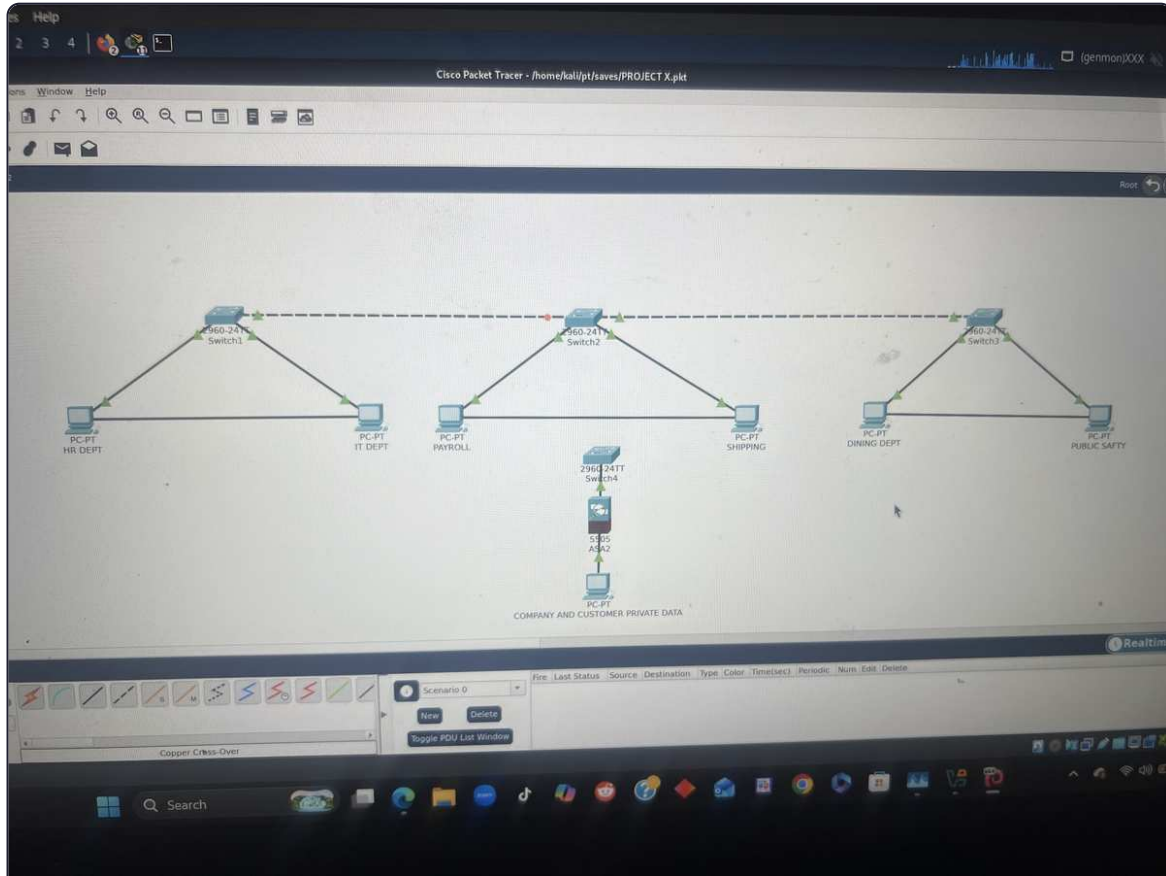
CDP warnings; broadcast leaking across VLANs. Fix: `switchport trunk native vlan` must match on both ends.

// Topology Rebuild – Flat Network Failure

The original flat topology connected access switches directly to the router with no core switch – creating trunk and routing conflicts at scale, requiring a full redesign.

COMPANION TOPOLOGY – PROJECT X

ASA5505 firewall staged between Switch4 and the private data segment.



Companion topology (PROJECT X) – ASA5505 firewall staged between Switch4 and the private data segment

SKILLS DEMONSTRATED

SKILL	WHAT IT PROVES
VLAN creation & naming	Broadcast domain segmentation
802.1Q trunk configuration	Multi-VLAN transport over single links
Access port assignment	Per-port traffic control at Layer 2
Router-on-a-Stick subinterfaces	Inter-VLAN routing without L3 switch
Hierarchical topology design	Scalability and enterprise architecture thinking
Subinterface/VID conflict diagnosis	CLI troubleshooting – not memorized config
Topology rebuild from failure	Engineering judgment, not just lab completion

PHASE 2 – SECURITY ENHANCEMENTS

☐ ACLs (least-privilege)	☐ Management VLAN 99
☐ Port Security	☐ DHCP Snooping
☐ BPDU Guard + PortFast	☐ SSH-only Management
☐ Disable Unused Ports	☐ DMZ VLAN
☐ Firewall (ASA5505)	☐ Syslog / SIEM
☐ DHCP Services	☐ STP Tuning

KEY TAKEAWAY

The Most Technically Valuable Part Was Diagnosing What Broke

The most technically valuable part of this project wasn't writing configs that worked – it was diagnosing why they didn't. Understanding that a subinterface needs encapsulation before accepting an IP, that trunks must match on both ends, that a flat topology breaks at scale – that diagnostic process is what network engineering actually looks like. Rebuilding a broken topology into a working hierarchical enterprise network is the real lab.

GENMONXXX / ENTERPRISE-CAMPUS-NETWORK-LAB

Cisco IOS 15.1 · Kali Linux 2025.2 · Cisco Packet Tracer